Hints: Euclidean geometry – 1. Vectors, angles, simplices and volumes in $\mathbb{R}^n$

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Realise the simplex as $A_i = \frac{a}{\sqrt{2}} e_i$ $(i = 0, \dots, n)$ in the hyperplane $x_0 + \cdots + x_n = \frac{a}{\sqrt{2}}$ of $\mathbb{R}^{n+1}$: all edges equal a, and the centre is $G = \frac{a}{\sqrt{2}} \cdot \frac{1}{n+1}\mathbf{1}$.

Hint 2. $|A_i - G|^2 = \frac{a^2}{2}\left(1 - \frac{1}{n+1}\right)$ gives R, and $\langle A_i - G, A_j - G\rangle = -\frac{a^2}{2(n+1)}$ gives the cosine $-\frac{1}{n}$. The centroid divides each median in the ratio $n : 1$, so the distance from G to a facet is $r = \frac{R}{n}$.

Hint 3. The Gram matrix of $A_i - A_0$ $(i \geqslant 1)$ is $\frac{a^2}{2}(E + J)$ with determinant $\left(\frac{a^2}{2}\right)^n (n+1)$; Theorem 1 gives the volume. The outer normal of the facet opposite A_i is parallel to $G - A_i$, so the outer normals of two facets make the angle $\arccos(-\frac{1}{n})$ and the dihedral angle is $\pi - \arccos(-\frac{1}{n}) = \arccos\frac{1}{n}$.

Problem 2

Hint 1. Expand $|XA_i|^2 = |(X - G) + (G - A_i)|^2$ and use $\sum_i (G - A_i) = 0$.

Hint 2. Put $X = O$: $(n+1)R^2 = (n+1)|OG|^2 + \sum_i |GA_i|^2$. Put $X = A_j$ and sum over j: $2\sum_{i<j} |A_iA_j|^2 = 2(n+1)\sum_i |GA_i|^2$.

Hint 3. Combine the two identities. Since $|OG|^2 \geqslant 0$, $\sum_{i<j} |A_iA_j|^2 \leqslant (n+1)^2 R^2$ with equality iff $O = G$; for a tetrahedron in the unit sphere the bound is 16.

Problem 3

Hint 1. Take the vertices of the icosahedron $(0, \pm 1, \pm\varphi)$ and their cyclic permutations, $\varphi = \frac{1+\sqrt{5}}{2}$. The six diagonals are spanned by $(0, 1, \pm\varphi)$, $(1, \pm\varphi, 0)$, $(\pm\varphi, 0, 1)$; all have $|v|^2 = 1 + \varphi^2 = \varphi + 2$, and every scalar product of two of them is $\pm\varphi$ or $\pm(1 - \varphi^2) = \mp\varphi$. Since $\frac{\varphi}{\varphi+2} = \frac{1}{\sqrt{5}}$, the lines are equiangular.

Hint 2. The 28 vectors have coordinate sum 0, so they lie in the hyperplane $\mathbf{1}^\perp \cong \mathbb{R}^7$, and $|v|^2 = 24$. Two of them with disjoint pairs of 3's have scalar product $-6 - 6 + 4 = -8$; with one common position, $9 - 3 - 3 + 5 = 8$. So all angles have cosine $\pm\frac{1}{3}$.

Hint 3. With $n = 2$ (three lines at $60°$), $n = 3$ (six lines) and $n = 7$ (28 lines) the bound $\frac{n(n+1)}{2}$ is attained.

Problem 4

Hint 1. Write the ellipsoid as $x^T Q x = 1$ with Q positive definite, and let $u_1, \dots, u_n$ be the orthonormal directions of OA_i, so $A_i = t_i u_i$ with $t_i^{-2} = u_i^T Q u_i$.

Hint 2. In the orthonormal basis u_i the hyperplane through $A_1, \dots, A_n$ is $\sum_i \frac{x_i}{t_i} = 1$, so its distance from O is $\left(\sum_i t_i^{-2}\right)^{-1/2}$.

Hint 3. $\sum_i u_i^T Q u_i = \operatorname{tr} Q$ for every orthonormal basis, so the distance is $(\operatorname{tr} Q)^{-1/2}$, independent of the choice.

Problem 5

Hint 1. Answer: $N = 3m^2$. Example: (m, m, m), $(m, -m, -m)$, $(-m, m, -m)$, $(-m, -m, m)$.

Hint 2. The centre of the tetrahedron is the centre of the sphere, so the vertices $v_1, \dots, v_4$ satisfy $\sum v_i = 0$, $|v_i|^2 = N$ and $\langle v_i, v_j \rangle = -\frac{N}{3}$. Look at the integer vectors $w_j = v_1 + v_j$, $j = 2, 3, 4$.

Hint 3. They are pairwise orthogonal and $|w_j|^2 = \frac{4N}{3} = L^2$. The integer matrix W with rows w_j has $WW^T = L^2 E$, so $\det W = \pm L^3$ is an integer; together with $L^2 \in \mathbb{Z}$ this gives $L \in \mathbb{Z}$. Finally $N = \frac{3}{4}L^2$ is an integer only for even L, so $N = 3\left(\frac{L}{2}\right)^2$.

Problem 6

Hint 1. Answer: exactly the odd n. For odd n take $J - E$: rows have $n - 1$ ones (even self-products), two rows share $n - 2$ ones (odd products).

Hint 2. For even n reduce modulo 2: $MM^T \equiv J - E$, and $\det(J - E) = (-1)^{n-1}(n - 1)$ is odd, so M is invertible over $\mathbb{F}_2$.

Hint 3. But over $\mathbb{F}_2$ we have $r \cdot r \equiv r \cdot \mathbf{1}$ for every row r, so every row is orthogonal to $\mathbf{1}$, i.e. $M\mathbf{1} \equiv 0$: a contradiction.

Problem 7

Hint 1. Four points of a plane form a degenerate 3-simplex, so the Cayley–Menger determinant of Theorem 1 (the bordered 5×5 matrix with the squared distances) vanishes.

Hint 2. The square of an odd integer is $\equiv 1 \pmod 8$. So modulo 8 the bordered matrix is $J - E$ of size 5.

Hint 3. $\det(J - E) = (-1)^4 \cdot 4 = 4 \not\equiv 0 \pmod 8$, a contradiction.

Problem 8

Hint 1. Let $P_i = v_i v_i^T$. Then $(v_i \cdot v_j)^2 = \operatorname{tr}(P_i P_j) = \langle P_i, P_j \rangle$ (the scalar product of matrices), $\langle P_i, E \rangle = 1$ and $\langle E, E \rangle = n$.

Hint 2. Hence $n(v_i \cdot v_j)^2 - 1 = n\langle P_i - \frac{E}{n}, P_j - \frac{E}{n} \rangle$: a Gram matrix, so positive semidefinite.

Hint 3. The sum of all entries of a positive semidefinite matrix is $\mathbf{1}^T G \mathbf{1} \geqslant 0$: $m(n-1) + m(m-1)(n\alpha^2 - 1) \geqslant 0$, which is the bound. For the examples: $n = 2, \alpha = \frac{1}{2}$ gives 3; $n = 3, \alpha = \frac{1}{\sqrt{5}}$ gives 6; $n = 7, \alpha = \frac{1}{3}$ gives 28.

Problem 9

Hint 1. Look for equilateral triangles with side $2\sqrt{2}$: three points that pairwise differ in exactly two coordinates. For instance, all neighbours of one point x of the cube (points differing from x in exactly one coordinate) pairwise differ in exactly two coordinates.

Hint 2. For $x \in \{\pm 1\}^n$ let $N(x)$ be its n neighbours. If $|B \cap N(x)| \geqslant 3$ for some x (not necessarily in B), we are done.

Hint 3. Double counting: every point of B lies in $N(x)$ for exactly n points x, so $\sum_x |B \cap N(x)| = n|B| > 2^{n+1} = 2 \cdot 2^n$, and some x has $|B \cap N(x)| \geqslant 3$.

Problem 10

Hint 1. Answer: $d(k, n) = kn$ if $k, n \geqslant 2$, and $d(k, n) = k + n$ if $k = 1$ or $n = 1$. For $k = 1$, "1-generic" means "not contained in a hyperplane", and $n + 1$ affinely independent points suffice.

For $n = 1$ the hyperplanes are points, a set is k-generic iff it has at least $k + 1$ points, and $k + 1$ points suffice.

Hint 2. Lower bound for $k, n \geqslant 2$: put k points on each of the n coordinate axes, away from the origin. A hyperplane not through 0 contains at most one point of each axis and misses at least $n(k-1) \geqslant k$ points; one through 0 cannot contain all axes and misses the k points of some axis. Removing any point p of axis i spoils this: the hyperplane through 0 containing all other axes misses only $k - 1$ points.

Hint 3. Upper bound: let $T \subseteq S$ be a minimal k-generic subset. Minimality gives, for every $p \in T$, a hyperplane $H_p \not\ni p$ that misses exactly k points of T. Build a chain: $p_1 \in T$, $L_1 = H_{p_1}$; then $p_{j+1} \in T \cap L_j$ and $L_{j+1} = L_j \cap H_{p_{j+1}}$. Since $p_{j+1} \in L_j \setminus L_{j+1}$, the dimension drops at every step, so after at most n steps $T \cap L$ has at most one point, and T is covered by at most n sets $T \setminus H_{p_j}$ of size k plus at most one point: $|T| \leqslant kn + 1$.

Hint 4. Suppose $|T| = kn + 1$. Then the n sets $C_j = T \setminus H_{p_j}$ are disjoint, and there is a point $q \in T$ lying in all H_{p_j} with $\bigcap H_{p_j} = \{q\}$. Every other point of T lies in exactly one C_j, i.e. in all H_{p_i}, $i \neq j$, which intersect in a line through q. So T is exactly the configuration of Hint 2 (with q as origin) plus the point q, and $T \setminus \{q\}$ is still k-generic, contradicting minimality.